

Hydrogen bonding enhances the electrostatic complex coacervation between κ -carrageenan and gelatin



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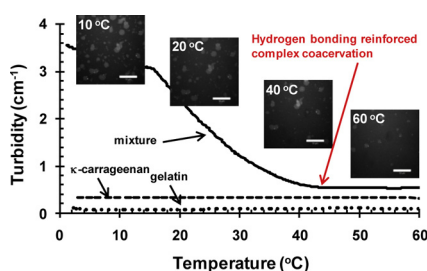
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HIGHLIGHTS

- κ -Carrageenan/gelatin mixture exhibited a peculiar turbidity increase on cooling.
- It is attributed to the hydrogen bonding formation between the biopolymers.
- Hydrogen bonding could reinforce their electrostatic complex coacervation.
- Effects of pH, ionic strength, mixing concentration and ratio were clarified.

GRAPHICAL ABSTRACT



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ABSTRACT

The phase behavior of sodium type κ -carrageenan (KC) and type B gelatin during cooling was studied using differential scanning calorimetry, thermal scanning rheology, turbidimetry, and confocal laser scanning microscopy. The turbidity of each single biopolymer solution and its mixtures stayed low above 40 °C, but that of the mixture began to increase steadily at 40 °C and then slowed down at 15 °C. Although it is already known that KC/gelatin mixture shows complex coacervation via electrostatic attraction, the dramatic increase of the turbidity below 40 °C is reported for the first time here. This peculiar transition is ascribed to the coacervation reinforced by hydrogen bonding between KC and gelatin. The second step of the turbidity increase below 15 °C is presumably related to the formation of KC helices that alters the hydrogen bonding with gelatin. The mechanism of hydrogen bonding reinforced complex coacervation was confirmed by mixing calorimetry and by examining the effects of urea and glycerol. Factors influencing the transition, including pH, ionic strength, mixing concentration and ratio, were discussed in relation to complex coacervation.

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1. Introduction

The interaction between biopolymers plays an important role in the structure, stability and rheological properties of food [1,2]. Understanding the interaction, especially the phase separation of biopolymer mixtures, is expected to improve food texture and stability [3,4].

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Phase separation can be classified into aggregative and segregative phase separation [5,6]. In general, segregative phase separation induced by thermodynamic incompatibility occurs at high ionic strengths and high polymer concentrations. The thermodynamic incompatibility of proteins and polysaccharides has been studied for more than hundred years [7]. Associative phase separation which is known as complex coacervation occurs at low ionic strengths and low polymer concentrations. Biopolymers with opposite electric charges attract each other and separate into two phases: one rich in both biopolymers and the other phase depleted in biopolymers and consisting mainly of solvent [8]. In more complex situation, three phases may also coexist [9]. Association has also been found for slightly negatively charged gelatin and KC mixture due to the presence of “patches”, positively charged regions, on protein [10,11]. Associative phase separations between polysaccharides and proteins have been widely investigated and characterized by turbidity measurement, light scattering and other physical measurements [12–14].

Antonov and Gonçalves [10] reported that the complexes were formed and stabilized via electrostatic interactions, rather than through hydrogen bonding or hydrophobic interaction by checking the dissociation effect of salts. In their work, the turbidity of gelatin/KC with high mixing ratio did not change on cooling from 50 °C to 20 °C, which was thought to be an evidence of insignificant influence of hydrogen bonding and hydrophobic interaction in complexation. However, there are some arguments that electrostatic interaction is an essential but not always a sufficient condition for the complex formation [10]. Intra- and inter-molecular hydrogen bonding, hydrophobic interactions can be affected by electrostatic interaction or other long range molecular forces [15].

It has been reported that extrinsic parameters such as pH, temperature, ionic strength have a great influence on phase behavior in addition to intrinsic parameters including molecular weight, molecular conformation, charge density, mixing ratio, and total polymer concentration [2,4,16].

κ -Carrageenan (KC) is extracted from red seaweeds [17]. It is an unbranched negatively charged hydrophilic polysaccharide with the repeating unit of disaccharide sequences of 4-sulfate esters of β -1,3-linked D-galactose and α -1,4-linked 3,6-D-anhydrogalactose. KC differs from ι - and λ -carrageenan in the content and position of sulfate groups and the content of the 3,6-anhydrogalactose residue. For the protein/carrageenan complexation, sulfate groups on carrageenan play a decisive role, and thus ι -carrageenan was found to have higher binding constant than KC [18].

Type B gelatin is extracted from bovine skin and has been widely used in the food industry and biomedical engineering [19]. Gelatin consists of about twenty amino acids which are involved in hydrogen bonding, and has a unique sequence of glycine–proline–hydroxyproline which plays an important role in the triple helix formation and its stabilisation. Both KC and gelatin show a cold-set thermo-reversible gelation [20].

Although the phase behavior of the mixture of gelatin and κ -carrageenan or ι -carrageenan has been studied [6,10,14,16,21], some ambiguities remain unclarified. The main objective of the present study is to clarify the role of hydrogen bonding in the associative phase separation governed by electrostatic interaction and to examine the effects of pH, ionic strength, temperature, polymer concentration and mixing ratio on the phase behavior. Biopolymers are particularly rich in electrostatic and hydrogen bonding interactions, resulting in different conformations under various experimental conditions. The present work would shed more light on how the coupling of different molecular interactions leads to complex phase behaviors of biopolymer mixtures.

2. Materials and methods

2.1. Materials

Type B gelatin (batch no. G9382), a lime-treated gelatin from bovine bone, was purchased from Sigma Co., Ltd. The number and weight average molar mass, $M_n = 86.5$ kDa and $M_w = 173$ kDa, were determined at 25 °C using a gel permeation chromatography–multiangle laser light scattering (GPC–MALLS) (Wyatt Technology Corporation, USA), with phosphate buffer solution (1/15 M, pH 7.0) as both solvent and eluent. The isoelectric point (IEP) of 5.1 was determined by zeta potential measurement using a Nano-ZS ZetaSizer (Malvern Instruments, U.K.).

KC sample (Gelcarin GP-911NF) was a gift from FMC Biopolymer. This KC sample contained various metal ions and was converted to sodium type by ion exchange using Amberlite IR-120 resin purchased from Sigma Co., Ltd. The resin was first converted to H^+ form by washing with 1 M HCl. 1.0 wt% KC solution was prepared and was dialyzed (M_w cutoff 8–14 kDa) against water for 24 h to remove any free salt. The H^+ -form resin and the dialyzed KC were mixed at a ratio of 1:2 and stirred for 3 h in an iced water bath to avoid acid degradation. KC in the H^+ form was separated from the resin, and was adjusted to pH 8.0 using 1 M NaOH to obtain the sodium type KC. The purified sample was then lyophilized on a Modulyod-230 Freeze Dryer (Thermo Electro Corporation, Germany).

After the conversion, the cation content of the sodium type KC was determined by atomic absorption spectrometer: 6.32% Na^+ , 0.067% K^+ , 0.0027% Mg^{2+} , 0.0083% Ca^{2+} . The number and weight average molar mass, $M_n = 389$ kDa and $M_w = 467$ kDa, were determined by GPC–MALLS with 0.1 M sodium iodide solution as both solvent and eluent.

The moisture contents after freeze-drying, 10.0% for KC and 12.2% for gelatin, were taken into account in all experiments. Concentrations of KC and gelatin solutions were given in weight percent. Milli-Q water was used throughout the experiments. Urea, glycerol, NaCl, NaOH, HCl used in this work were all of analytical grade.

2.2. Preparation of solutions

All solutions were prepared in a vial by magnetic stirrer. Gelatin granules were dissolved to obtain a 1.5% aqueous solution in NaCl of various molar concentrations for 1 h at 60 °C. The dissolution temperature was kept lower than 60 °C to avoid the hydrolysis of gelatin chains. The pH of the gelatin aqueous solution was adjusted by the addition of 0.5 M NaOH or 0.5 M HCl at 60 °C. An aqueous solution of 1.5% sodium type KC was made by dissolving the sample into NaCl of various molar concentrations for 1 h at 85 °C. KC and gelatin solutions were blended at 85 °C under mild magnetic stirring for 10 min to obtain the desired concentrations. To prevent microorganism growth, 0.03% NaN_3 was added to each solution.

2.3. Differential scanning calorimetry (DSC)

2.3.1. Characterization of conformational transition

To investigate the conformational orderings of KC and gelatin molecules, a micro DSC III Evo (Setaram Instrumentation, France) was used. Each sample of 700 mg was loaded to a high pressure hastelloy cell and an equal amount of aqueous NaCl solution was used as the reference. The temperature was raised from room temperature to 70 °C at 3.0 °C/min and was held at 70 °C for 10 min, and then cooled to 0 °C at 0.5 °C/min, followed by heating again to 70 °C at 0.5 °C/min. Scans were done in two cooling–heating cycles to check the reproducibility.

2.3.2. Characterization of the interaction during complexation

To understand the interactions of KC and gelatin system at pH 7.0 and 100 mM NaCl, mixing calorimetry was performed in an isothermal mode at different temperatures [22]. Each 200 mg of 1.5% gelatin solution and 1.5% KC solution was loaded separately in the upper and lower compartments of a mixing cell. The released or absorbed heat was measured, relative to a reference cell containing the same amounts of the gelatin solution in the upper compartments and 100 mM NaCl solution in the lower compartment. The solutions in the upper and lower compartments were rapidly mixed, and then the heat flow with lapse of time was recorded. The obtained calorimetric curve was integrated, yielding a mixing enthalpy (mJ/g). All the measurements were done at least in triplicate.

2.4. Rheological measurements

Temperature dependence of storage and loss moduli, G' and G'' , of KC, gelatin and their mixtures were observed on cooling from 70 °C to 10 °C at 0.5 °C/min, and at a fixed frequency of 1 Hz and a stress of 0.5 Pa using a Haake RheoStress 6000 (Thermo Fisher Scientific, USA) equipped with a circulating bath AC 200 (Thermo scientific). A serrated parallel plate geometry (35 mm in diameter and 1 mm in gap) was used. The periphery of the solution was covered with a thin layer of silicone oil to prevent evaporation.

2.5. Turbidity measurements

Turbidity measurements were performed on cooling from 70 °C to 0 °C at 0.5 °C/min using a TU-1900 UV/vis Spectrophotometer (Persee, China) equipped with a temperature-controlling cell holder under nitrogen gas. The wavelength used was 500 nm, and the optical path length was 1 cm. The temperature was controlled by a Peltier device within ± 0.1 °C.

2.6. Confocal laser scanning microscopy (CLSM)

Microstructures of mixed KC/gelatin aqueous solutions were observed on a Zeiss LSM 510 META (Carl Zeiss AG, Germany) inverted confocal laser scanning microscope, equipped with a multiline argon laser excited at 547 nm. Before mixing, gelatin was stained using a small amount of 0.02% rhodamine B overnight at ambient temperature [4,16].

3. Results and discussion

3.1. Phase behaviours of individual biopolymer solutions and their mixtures

Thermal, rheological and optical behaviors for KC, gelatin and their mixtures in 100 mM NaCl were investigated. DSC curves of individual KC and gelatin in Fig. 1a both show a single exothermic peak which is attributed to the coil to helix transition of each biopolymer. While the exothermic peak for KC is sharp, that for gelatin is broader, which might be due to the polydispersity of the gelatin sample. It is well-known that longer chains begin to form helices at higher temperatures than shorter chains [20]. A main exothermic peak was observed for the mixture, which is similar to that of KC at 12 °C but shifts slightly to a higher temperature (14 °C). This might be caused by the polyelectrolyte effect of gelatin or by the increase in the effective concentration of KC.

In addition to this main exothermic peak at 14 °C, the thermogram for the mixture shows an exothermic shoulder at the temperature range from 14 to 20 °C. The shoulder should come from the ordering of gelatin chain, but it is smaller than that observed

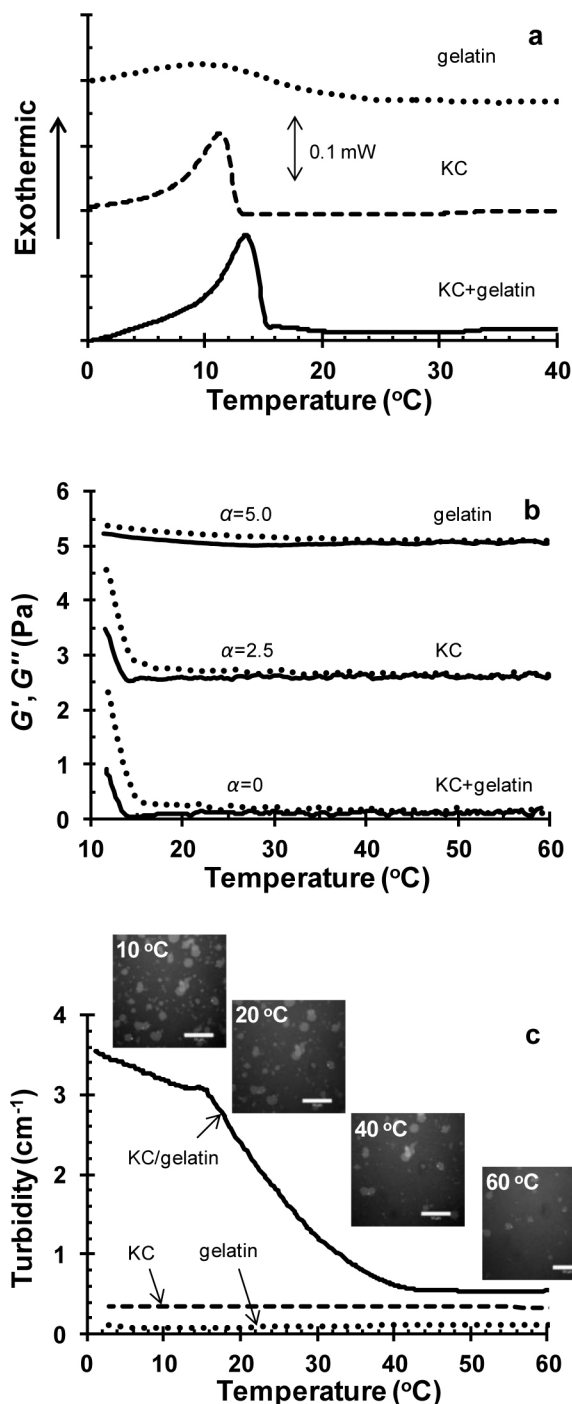


Fig. 1. (a) DSC curves, (b) storage and loss moduli (G' , solid line; G'' , dotted line), and (c) turbidity at 500 nm for 0.75% KC, 0.75% gelatin, and a mixture of 0.75% KC/0.75% gelatin during cooling at 0.5 °C/min. The data for the moduli are shifted along the vertical axis by a factor α to avoid overlapping. The corresponding CLSM images for the mixture at 10, 20, 40, 60 °C are shown as insets in (c). pH 7.0; NaCl = 100 mM; Scale bar = 50 μm.

for gelatin alone, indicating that structure ordering of gelatin is impeded by the coexisting KC.

Fig. 1b shows the temperature dependence of G' and G'' for KC, gelatin and their mixture on cooling. While G' and G'' show a steep increase at 14 °C for KC, those for gelatin show only a slight and gradual increase with lowering temperature. This increase in both moduli should be attributed to the coil-to-helix transition and not to the sol-to-gel transition because $G' < G''$ at all the temperatures

examined. A similar steep increase of G'' in non-gelling condition was ascribed to a coil-to-helix transition in gellan [23]. Comparison between KC, gelatin, and their mixture indicates that the rheological behavior of the mixture is mainly dominated by KC.

Although each individual biopolymer solution is transparent over the whole temperature range with no significant change in turbidity during cooling, the turbidity of mixed solution begins to increase at 40 °C (Fig. 1c). This increase is ascribed to the appearance of structural inhomogeneity [6,10]. The temperature at which the turbidity begins to increase is much higher than the onset temperature of ordering for individual biopolymers detected by cooling DSC (Fig. 1a) and by thermal scanning rheology (Fig. 1b), indicating that the structural inhomogeneity is not induced by the conformational transition from coil to helix. It was reported that at low ionic strength, KC and gelatin showed associative phase separation due to the electrostatic interaction [6,10,14,16]. From the corresponding CLSM images, it can be seen that the coacervates grow significantly below 40 °C (Fig. 1c), which coincides well with the turbidity increase. It should be pointed out that the transition at 40 °C upon cooling is completely thermo-reversible, and the subsequent heating can restore the turbidity and microstructure observed originally (data not shown).

The above results indicate that a structural change occurs in the mixed solution during cooling around 40 °C, which is much higher than the temperature of the conformational transition from random coil to helix for KC and gelatin. At temperatures higher than 40 °C, disordered KC and gelatin coexist and attract each other by opposite charges. The attraction of negatively charged gelatin and KC is attributed to the presence of positively charged “patches” in gelatin [16,21]. The dramatic increase of the turbidity below 40 °C is attributed to the transition from the associative phase separation dominated by electrostatic attraction to the one reinforced by hydrogen bond, as will be discussed later. Below 15 °C, the turbidity increase turns to slow down gradually. In this temperature range, KC molecules transform into helical state, as detected by DSC. Since individual KC solution shows no turbidity change in this temperature range, the second step of turbidity increase cannot be simply ascribed to the conformational transition of KC. Instead, it could be due to the alteration of hydrogen bonding between KC and gelatin when KC forms double helices.

3.2. Hydrogen bonding between KC and gelatin

For further understanding the interaction of KC and gelatin, mixing isothermal calorimetry was performed at different temperatures. Fig. 2a shows the heat flow as a function of time after mixing of 0.75% KC/0.75% gelatin. The absolute enthalpy value of mixing at each temperature determined from the area enclosed by the curve and the baseline is shown in Fig. 2b. At temperatures higher than 40 °C, the mixing enthalpy is relatively low, which is contributed mainly by electrostatic complex coacervation. The mixing enthalpy becomes increasingly negative (exothermic) when the temperature is below 40 °C. It suggests that a second mechanism comes into play and reinforces the complexation between KC and gelatin. The transition of the mixing enthalpy is in line with the turbidity increase described above.

To confirm the nature of the molecular interaction that underlies the transition, the effects of urea and glycerol on the transition were investigated. Urea is known as a hydrogen bond breaker [15,24]. Glycerol, as a hydrogen donor, can decrease biopolymer collision rate and increase biopolymer aggregation [25], and is thought to promote hydrogen bond formation. As shown in Fig. 3, the onset temperature of turbidity increase T_0 decreases strongly with increasing urea concentration. Actually, the transition is completely suppressed at 2 M urea and no turbidity increase can be observed (data not shown). On the contrary, the addition of glycerol

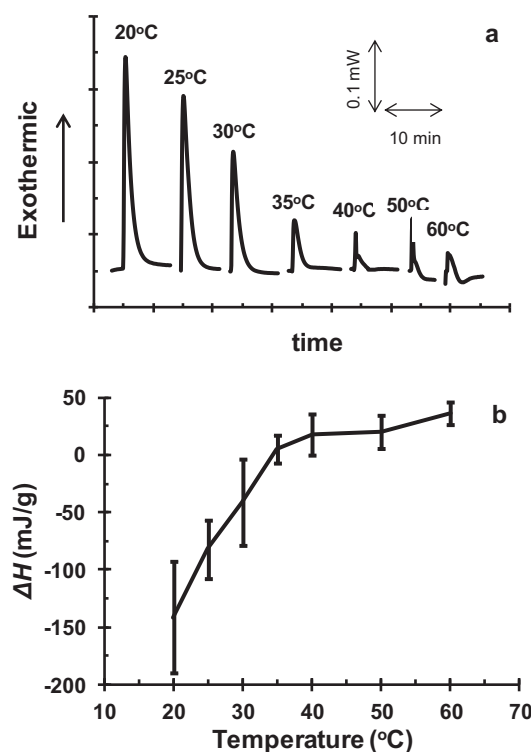


Fig. 2. (a) Calorimetric peaks for mixing 0.75% KC/0.75% gelatin mixtures at pH 7.0 and different temperatures in 100 mM NaCl. (b) Mixing enthalpy ΔH estimated from the integration of the calorimetric peaks at each temperature.

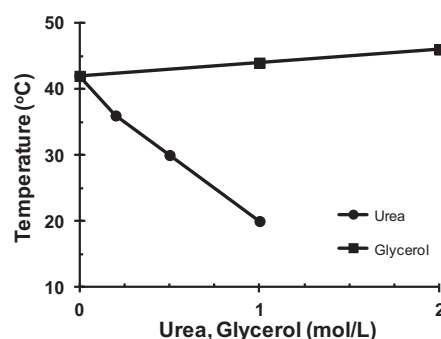


Fig. 3. The onset temperature T_0 of the turbidity increase as influenced by the addition of urea (circle) and glycerol (square) for 0.75% KC/0.75% gelatin mixtures in 100 mM NaCl at pH 7.0.

slightly increases T_0 . These results indicate that urea can greatly inhibit the transition while glycerol can promote it. It manifests that the transition around 40 °C for KC/gelatin mixture during cooling is most likely to originate from hydrogen bonding between KC and gelatin. Since KC and gelatin attract each other by electrostatic interaction as mentioned above, the distance between disordered chains of these two polymers might be shortened by the electrostatic attraction to favor hydrogen bonding formation. Since the transition is well above the conformational ordering temperatures of KC and gelatin, it could be possible that the local structural arrangement of the biopolymers or the change in local solvent environment prior to their conformational transitions enables the formation of intermolecular hydrogen bonds. Indeed, hydrogen bonding is also the major molecular force that stabilizes gelatin and KC helices in the ordered state [20].

KC and gelatin exhibit conventional complex coacervation due to electrostatic interaction. The electrostatic interaction is less temperature dependent than hydrogen bonding and hydrophobic

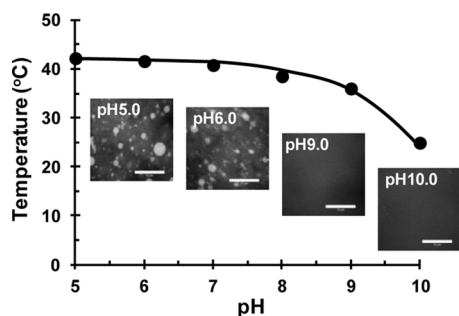


Fig. 4. The onset temperature of turbidity increase T_0 as a function of pH for 0.75% KC/0.75% gelatin in 100 mM NaCl. The insets are CLSM images for the mixtures with different pHs and taken at 30 °C. Scale bar represents 50 μ m.

interaction [15], and therefore the electrostatic complex coacervate shows a nearly constant turbidity in the temperature range of >40 °C (Fig. 1c). When the temperature is below 40 °C, intermolecular hydrogen bonding between KC and gelatin comes into play, which as an attractive interaction could reinforce the existing complex coacervation induced electrostatically. This leads to an extensive turbidity increase, as observed in Fig. 1c.

3.3. Factors influencing the transition

3.3.1. Effect of pH

The change in pH has been known to influence the complex coacervation of protein–polysaccharide mixture [2]. It was shown previously that pH influenced the associative phase separation boundary and markedly changed the state diagrams of KC/gelatin mixture [6]. With increasing pH, the onset temperature of the turbidity increase T_0 shifts to lower temperatures (Fig. 4). This means that the increase in pH unfavors the hydrogen bonding reinforced complex coacervation. The inset CLSM images show that the microstructure of the complex coacervate at 30 °C is greatly influenced by pH. The coacervate phase is reduced with increasing pH and nearly disappears at pHs >9.0.

At higher pHs, the electrostatic attraction between gelatin and KC is much weakened since most of cationic groups in gelatin are deprotonated. This leads to the dissociation of electrostatic complexes between KC and gelatin [6,10]. It seems that the reduction in electrostatic complexes makes it difficult for the subsequent formation of hydrogen bonding between KC and gelatin, as reflected by the shift of T_0 toward lower temperatures. A possible explanation is that the dissociation of electrostatic complexes increases the spatial distance between KC/gelatin and even prevents them from closely contacting due to electrostatic repulsion, thus unfavoring intermolecular hydrogen bonding.

3.3.2. Effect of ionic strength

For individual KC and gelatin, chains are expanded by intramolecular electrostatic repulsion between charges of the same sign. Addition of salt is generally believed to shield the repulsion, and thus makes the expanded chains into contracted coils. In the state of complex coacervate, KC and gelatin entangle each other by electrostatic attraction between the sulfate groups of KC and the positive patches of gelatin, and the effect of the addition of salt is quite delicate. It has been well established that there is an optimal concentration of salt for the formation of electrostatic complexes between charged polysaccharide and protein [26]. At lower salt concentrations, usually within several ten millimoles per liter depending on the valency of the salt, the addition of salt promotes the electrostatic complex coacervation. This has been explained as follows: (1) the addition of a small amount of salt makes biopolymer chain less rigid by screening intramolecular

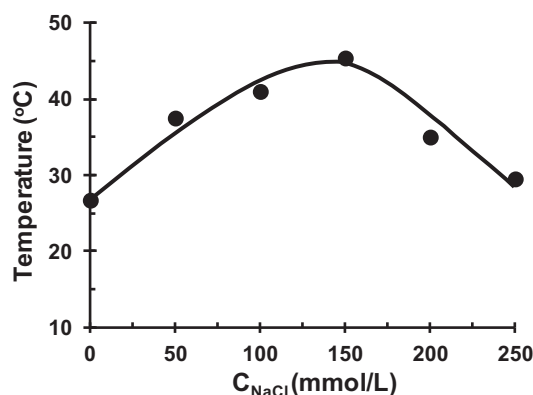


Fig. 5. The onset temperature of turbidity increase T_0 vs. NaCl concentration for 0.75% KC/0.75% gelatin in 100 mM NaCl at pH 7.0.

repulsion, and favors the occurrence of complex coacervation [6]; (2) the addition of a small amount of salt shortens the Debye length and can screen long-range intermolecular electrostatic repulsion between similarly charged biopolymer segments or patches, thus facilitating short-range intermolecular electrostatic attraction and consequently the complex coacervation [26]. However, if excessive amount of salt is added, the Debye length is greatly reduced and it leads even to the screening of the short-range electrostatic attraction, eventually suppressing the complex coacervation.

As can be seen from Fig. 5, the onset temperature T_0 for the hydrogen bonding reinforced complex coacervation indeed exhibits a maximum transition temperature around NaCl = 150 mM. At the lower salt side, increasing NaCl concentration increases T_0 , indicating that the formation of hydrogen bonds is promoted. At the higher salt side, further addition of salt decreases T_0 , indicating that the formation of hydrogen bonds is suppressed. Therefore, the effect of salt on hydrogen bonding formation between KC and gelatin is in concert with the effect on electrostatic complexation. Electrostatic complexation seems to be an advantage for the subsequent formation of hydrogen bonding between KC and gelatin.

However, the optimal concentration of NaCl observed in Fig. 5 (150 mmol/L) is somewhat larger than the usually expected value of several ten millimoles per liter for monovalent salts [6,26]. This cannot be explained only in terms of electrostatic complexation. Additional effect of NaCl on solvent quality, i.e., the hydration of the biopolymer, should also be taken into consideration. Increasing NaCl concentration might decrease the hydration of KC and gelatin, and promote the intermolecular hydrogen bonding between them.

3.3.3. Effect of mixing concentration and ratio

The onset temperature T_0 of the transition at a fixed KC/gelatin ratio, as a function of total biopolymer concentration, is shown in Fig. 6a. T_0 gradually decreases as the concentration decreases. Below the concentration of 0.4%, the complex coacervate could not be visually observed at 30 °C under CLSM. With decreasing biopolymer concentration, long-range electrostatic repulsion becomes more important, and the electrostatic attraction between KC and gelatin is weakened as well as the hydrogen bonding. It again suggests that hydrogen bond interaction is closely related with the electrostatic interaction. The observation that high ionic strength, high pH, and low concentration all lower the T_0 demonstrates that the transition from the associative phase separation dominated by electrostatic interaction to the one reinforced by hydrogen bonding should occur on the premise of the existence of electrostatic interaction (Figs. 4–6).

T_0 as a function of mixing ratio at a fixed total concentration of biopolymers is shown in Fig. 6. T_0 exhibits a clear sigmoidal change

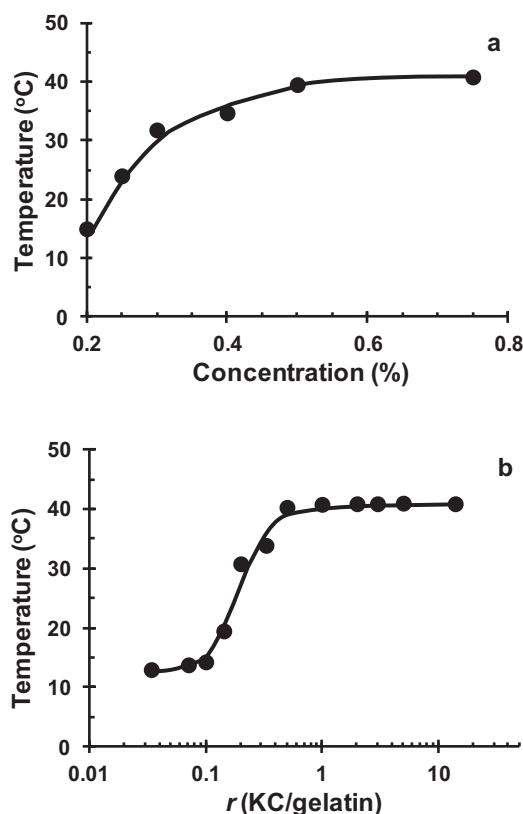


Fig. 6. The onset temperature of turbidity increase T_0 as a function of total biopolymer concentration at a fixed KC/gelatin ratio of $r=1$ (a) and as a function of mixing ratio r while at a fixed total biopolymer concentration of 1.50% (b). pH 7.0; NaCl = 100 mM.

with increasing mixing ratio r , and begins to increase markedly at $r=0.2$. The change is very similar to that observed for isoelectric point as a function of r [26]. The ratio of 0.2 might indicate a stoichiometry of the complexation of gelatin with KC. At mixing ratios higher than 0.2, KC molecules are in excess and intramolecular complexes are mainly formed [26]. This could imply a maximum contacting between KC and gelatin, and facilitates the formation of hydrogen bonds. On the contrary, when $r < 0.2$, gelatin molecules are in excess and intermolecular complexes bridged by gelatin are formed [26]. This might prevent the side-by-side alignment of KC and gelatin chains and thus impedes the formation of extensive hydrogen bonds (or hydrogen bonding network). Therefore, the effect of mixing ratio on T_0 should also be explained in the context of electrostatic complexation.

4. Conclusions

It was shown that a 0.75% KC/0.75% gelatin mixed system underwent a transition from complex coacervation governed by electrostatic interaction to that reinforced by hydrogen bonding upon cooling. The increasing of turbidity at 40°C is not induced by the ordering of two polymers both of which are still in coiled state. Urea and glycerol have been used to identify the driving force in this transition. The transition is found to be mainly due to the hydrogen bond formation between KC and gelatin that reinforces the existing complex coacervation. pH, ionic strength, mixing concentration and ratio are the major parameters that affect the hydrogen bonding reinforced complex coacervation. Their effects should be explained in the context of electrostatic complexation. Electrostatic complexation seems to be an advantage for the subsequent formation of hydrogen bonding upon cooling. The study would provide

a better understanding on how the coupling of different molecular interactions that is rather common in biopolymer mixtures determines their complex phase behaviors.

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